

Answer to the Editor (to be compiled in a pdf after removing underlined text or including it in the new text added):

Dear Editor,

We would like to thank the referees for reviewing this paper and furnishing this report.

We have carefully considered all comments, and we have applied some changes to the original version of the paper to address the issues raised. Detailed responses to all the comments can be found below (responses of the authors in blue, modified text in green, with additions specifically highlighted in bold, and parts that were completely removed in strikethrough red).

We remain at your disposal for any further clarifications and/or additional information.

Sincerely,
ATLAS Collaboration

Referee #1:

Dear authors,

Thank you for this nicely well-written paper. The manuscript is well-written, the results are interesting and I believe the paper deserves to be published after some minor revisions. Please find my detailed comments below.

Introduction:

In the SM, the production of a $t\nu$ system with an associated n -quark is SM suppressed. ->I feel the statement is too generic. There are SM processes (e.g. top quark decay) where there is no issue. So it would be good to be more precise here. I assume it is referring to b -meson decays.

You are right, we were referring to the process $gq \rightarrow Wb \rightarrow t\nu b$, rather than to all SM processes yielding a $t\nu + b$ final state. This process is CKM-suppressed by the small V_{ub} and V_{cb} . The sentence was rewritten accordingly.

Old text: "In the SM, the production of a $t\nu$ system with an associated n -quark is SM suppressed."

Revised text: "In the SM, the production **amplitudes of a $t\nu$ system with an associated b -quark, which can directly interfere with the BSM signal, are**

CKM-suppressed. Consequently, the SM-BSM interference is expected to be small in the high-mass region.”

Section 3

Signal model footnote. This is a bit confusing to me. You say that the β_{13} is roughly $0.2\beta_{23}$ and then you say it is ignored. Does this mean that you actually put it to 0 in the end?

The $\beta_{L^{13}}$ parameter is included in the generation of the signal samples and is set to $V_{td}/V_{ts} \cdot \beta_{L^{23}}$. It is only omitted from Eq. (1) since it has a negligible impact of $\left| \frac{V_{cd} \cdot \beta_{L^{13}}}{V_{cs} \cdot \beta_{L^{23}}} \right| \sim 0.05$ on the final C_{LL} coefficient.

SK – Modification suggested for footnote to make it more clear:

Old footnote: “The parameter $\beta_{L^{13}}$ is set to $V_{td}^*/V_{ts}^* \cdot \beta_{L^{23}}$ following the convention of Ref. [33]. Its contribution is smaller than the $\beta_{L^{23}}$ term and is ignored here for simplicity.”

Revised footnote: “The parameter $\beta_{L^{13}}$ is set to $V_{td}^*/V_{ts}^* \cdot \beta_{L^{23}}$ following the convention of Ref. [33]. Its contribution to Eq. (1) is approximately 5% of the $\beta_{L^{23}}$ corresponding term and is neglected there for simplicity.”

Section 5

Why is there no muon veto for hadronic tau candidates?

The probability for a muon to be misidentified as a τ_{had} candidate is much smaller than for an electron. Therefore, only the electron-veto RNN is applied to the τ_{had} candidates.

Define a calorimeter-tagged muon before

We have adjusted the sentence adding a definition for a calorimeter-tagged muon.

Revised text: “calorimeter-tagged muons (identified from ID tracks matched with a calorimeter energy deposit compatible with a minimum ionizing particle) sharing an ID track with an electron”

Why is for an electron only the closest jet vetoes, while for muons all jets close by are vetoed if they have max. 2 tracks?

Not all jets close to a muon are removed. A jet is removed only if the muon is ghost-associated with it within $\Delta R=0.4$ and the jet has at most two associated tracks.

Electrons typically produce sizable calorimeter showers, while muons are usually minimum-ionising. This motivates the different strategies for jet-electron and jet-muon overlap removal. We have revised the wording accordingly.

Revised text: “jets within $\Delta R=0.4$ of a muon if they have at most two associated tracks **and the muon is ghost-associated with the jet**”

ETmiss is redefined here while it was defined before

We have removed the relevant part that redefines it.

~~“The magnitude of this vector is denoted by ETmiss.”~~

Event selection

In general it could use some more motivation for some of the cuts. Now they are mostly listed. It is nice to connect the requirements to the background which is being rejected (e.g. why max. 4 jets). What kind of criterion was used? I see that the stat. unc. on bkg is kept small but assume there was also a specific criterion used to extract signal?

The upper bounds on the jet multiplicity suppress the top quark background. The high jet- p_{T} and the invariant mass requirements are used to select events consistent with an on-shell LQ decay (resonant LQ production), whereas the large m_{T} , $E_{\mathrm{T}}^{\mathrm{miss}}$ and $\Delta\phi$ requirements target the high-mass, back-to-back taunu topology for the non-resonant LQ process. The specific value for each selection was determined through the SR optimisation by scanning the selection thresholds and maximising the expected significance, while requiring the statistical uncertainty on the total background to remain below 30%.

SK – Sentence(s) that could be added (one or both):

Old text: During the optimisation of the SRs, using simulated events, the selections are chosen such that the statistical uncertainty of the expected background is kept below 30%.

Revised text: “During the optimisation of the SRs, using simulated events, the selection **thresholds were chosen by maximising the expected exclusion sensitivity, while requiring the MC statistical uncertainty in the expected background to remain below 30%.**”

and/or

New text: “**The jet multiplicity requirements suppress top-quark background, while the remaining kinematic requirements target the resonant or non-resonant signal topology.**”

For the m_{thad} the light jet is a proxy for the b-quark, otherwise it could not give the leptoquark mass. What is the purity of this selection? Have you checked that it would not be better to use the highest GN-2 value instead? Maybe that would have more purity?

The leading jet in SR0b-Res is not selected simply as a proxy for a b-quark. For large β_L^{23} values, the SR0b-Res has a sizeable contribution from resonant $U_1 \rightarrow s\tau$ decay. In particular, $U_1 \rightarrow$ charged lepton decay width for $b\tau:s\tau:d\tau$ is approximately $1:|\beta_L^{23}|^2:|\beta_L^{13}|^2$, which means that fraction of $s\tau$ decay increase from about 3.8% to about 74%, when β_L^{23} increases from 0.2 to 1.8. Selecting the light jets with the highest GN2 score is therefore not appropriate in the SR0b-Res region.

SK: Add a relevant sentence in Sec4? Again probably not so necessary, I have not added something for this one

Background estimation

Why are we only considering fully leptonic $t\bar{t}$ and tW decays? Explain why this is dominant. Also, we estimate both together, but they must behave quite a bit different in different bins and also in the extrapolation? How is that dealt with? Do you have a ratio uncertainty for them?

We studied the inclusive $t\bar{t}$ and single top production and confirmed that the leptonic $t\bar{t}$ and tW decays constitute the majority of top background after the large $E_{\text{T}}^{\text{miss}}$ requirement. The single-top/ $t\bar{t}$ ratio was also compared between the SRs and CRs and found to be consistent within the MC statistical uncertainties.

I would swap $t\bar{t}$ and W-jets section to be consistent with the discussion before.

We kept the same ordering throughout the draft, with the W+jets discussion preceding the top-quark discussion. The same order is used in the summary plot.

For the W+jets section, in the final hadronic tau case, there will be extra neutrino's wrt leptonic case? How do you make sure this does not mess up your modeling?

Indeed there is the extra neutrino in the hadronic tau case, but the $W \rightarrow e\mu\nu$ to $W \rightarrow \tau\nu$ extrapolation has been validated in the low-mass/low- m_{T} $W \rightarrow \tau\nu$ region, with good agreement between data and simulation.

Systematic uncertainties

30% uncertainty on W+HF vs. W+LF. Was a cross-check made for this analysis too? Why would these old ATLAS studies be valid for this analysis, in another phase space. It would be good to justify this.

As shown in Refs. 87-90, indeed the W+HF/W+LF mismodeling depends on the event kinematics. The 30% uncertainty is chosen conservatively to cover the largest deviations observed in the previous ATLAS studies.

Since the analysis is statistically dominant, further increasing this uncertainty has a negligible impact on the final result. This uncertainty is also decorrelated between the resonant and non-resonant regions to account for possible kinematic dependence.

For the $Z \rightarrow \nu\nu$ background, why was the choice made not to use a fake tau method, like almost all similar searches do? We know that the modeling is bad and it is confirmed by the 60% differences you are already seeing. Would it not be better to correct using such a method? Now we apply a 100% uncertainty on a MC method. The 100% is also not really motivated.

The contribution from the fake- τ background in this analysis is negligibly small. Even a 100% uncertainty changes the final result by less than 1%. For this reason, a dedicated fake- τ method is not needed for this background. The 100% uncertainty is assigned conservatively and should cover the differences up to 60% observed in the fake study.

Results

Motivate why you completely uncorrelated the W normalization (same for top) in the two regions. Connected to this. For top the two factors are roughly similar, but for the W they are very different. Do you understand why the W normalization in both regions would be so different?

The normalizations are decorrelated between the Res and Non-Res regions because they probe different kinematic regimes. The difference between the fitted W+jets normalization factors can be explained as a result of the different jet- p_{T} and $E_{\mathrm{T}}^{\mathrm{miss}}$ ranges in the two regions.

It is mentioned that it is verified that the total yield stays positive in all cases, but is this also the case during the limit setting when μ is allowed to vary or could weird things happen there? Do you constrain how much μ can vary?

In general, the contribution from BSM+interference could become negative in cases where μ is small enough. However, the contribution from interference is small in the parameter region to which the analysis is sensitive, and the upper-limit determination is not affected.

Discrepancy in Validation Region. For this discrepancy, could it be related to the big difference in normalization factors? If there is a specific distribution that convinced you that it is a statistical fluctuation, consider adding it to the supplementary material

The $E_{\mathrm{T}}^{\mathrm{miss}}$ distribution in that region and a dedicated check in a lower mass VR indicate that the discrepancy is localized to a single bin and is consistent with a statistical fluctuation. Since the supplementary material is pretty packed already, we prefer not to add that plot, but we would like to share it with the referees in this response.

SK: Share internal plot with the referee. @Jiaqi can you send it to us in pdf format if possible?

I am also missing a bit of information about the correlation model of the nuisances. Which ones are correlated between signal and background, across bins, across years?

The nuisance parameters are generally correlated across regions and between signal and backgrounds, where applicable. Theoretical uncertainties are process-specific, while W+jets HF modeling uncertainty is decorrelated between the resonant and non-resonant regions. The fit uses one bin per region, and no special treatment is applied across the years.

SK – Sentences that could be added:

New text (Section 8): “The $W\ell$ HF modelling uncertainty is decorrelated between the resonant and non-resonant regions.”

New text (Section 9): “Nuisance parameters are correlated across analysis regions and between signal and background samples, where applicable, unless stated otherwise.”

Referee #2:

First of all, congratulations on this interesting result by ATLAS. The overall quality of the document is really good. I have some comments that will help you improve your paper. Please find them below:

page 2 last paragraph: Two types...

In the resonant and non-resonant categories, do you consider only the signal that each category is dedicated to?

Similarly, for the b-tagging categories. What is considered as signal in each category and what do you do for the other signals? Is each signal extracted from the dedicated category?

We do not separate the signal according to the analysis categories. The resonant and non-resonant SRs are optimised for the corresponding topologies, while both can contain contributions from the other topology (although usually much smaller). The same applies to the b-tagged and b-vetoed categories. All four SRs are used simultaneously in the signal extraction.

page 4 4th paragraph:

How reasonable is it to look only in the left handed scenario?

The pure left-handed scenario is chosen as the main benchmark since it directly probes the $b \rightarrow c \tau \nu$ transition motivating this search. We have also studied the

alternative benchmark with $|\beta_R| = 1$, as shown in the supplementary material.

We find that the right-handed coupling enhances the relative non-resonant contribution, leading to improved sensitivity at high LQ masses. We therefore consider the supplementary study sufficient to illustrate the impact of this alternative benchmark, while keeping the main interpretation focused on the purely left-handed scenario.

The upper limit on the combined resonant and non-resonant visible cross section therefore shows the dependence on the LQ mass. We could add a short description of this study in the text, something like “The impact of a sizable right-handed coupling is also investigated and is found to enhance the non-resonant process relative to the resonant one, resulting in stronger sensitivity at high LQ masses.”

page 4 last paragraph/ page 5 first: The relative contributions of...

Can you explain more the interference between resonant and non resonant signal and how this is taken into account in the signal extraction.

All resonant and non-resonant signal diagrams are included simultaneously in the matrix-element calculation. Therefore, any interference between resonant and non-resonant diagrams is naturally included in the nominal BSM samples, and no specific treatment is needed in the signal extraction.

page 5 second paragraph: no hadronic tau triggers? a bit surprising, you may want to comment.

The $E_{\text{T}}^{\text{miss}}$ trigger is already fully efficient for the signal-region selection, since all SRs require missing $E_{\text{T}}^{\text{miss}} > 200$ GeV. Therefore, no dedicated hadronic tau trigger is needed for the signal efficiency. Single-lepton triggers are used for the CRs and VRs.

SK – Sentence that could be added:

New text: “The $E_{\text{T}}^{\text{miss}}$ trigger is fully efficient above the offline $E_{\text{T}}^{\text{miss}}$ threshold used to select signal events. Therefore, no dedicated τ_{had} trigger is required.”

page 5 third paragraph: The contribution from interference between the signal process and the SM ...

Can you explain more this interference as well? Later on (page 11 in the result) you mention that the μ is set to $\sqrt{\mu}$, it is not obvious why.

The interference occurs between the BSM and SM τ_{nu} amplitudes.

Separate samples are used for the pure BSM term (S) and the interference term (I), which scale with μ and $\sqrt{\mu}$ in the fit, respectively. The expected yield is therefore written as $(\mu * S + \sqrt{\mu} * I + B)$ in the likelihood.

page 6 and page 9 second paragraph:

The names of the working points don't mean much to those of us out of ATLAS. This is especially true on page 9. Either numerical description of the efficiency or simply the mention that a "tight" selection is applied to ID and Iso is enough. My understanding is that the preselection leptons are used for the overlap removal, while the CR is tighter. It seems to me that this is fairly standard practice.

If there is another point that I have missed, and requires the exact names? If so make it more clear because as an educated reader I didn't get it.

Indeed your understanding is correct. The selections described in Section 5 are used for overlap removal and lepton vetoes, while tighter identification and isolation requirements are applied in the CRs. The names of the working points are defined in the sources cited that's why they were used as is.

SK – Sentence that could be added (although I don't find it necessary):

New text: **"The electrons and muons selected in the control regions are required to satisfy tighter identification and isolation criteria than those used in the region definition and rejection of overlapping events, which are defined in Section~\ref{sec:object}."**

page 11, results, first paragraph:

The interference is not very well described, appears almost out of context here. It is implied earlier but not enough information is given. also this is interference between the signal and the SM bkg right?

Is there also interference between resonant and non-resonant signal?

Why $\sqrt{\mu}$?

This is addressed above. We have also made the S+I parameterisation explicit in the Results section.

SK – Sentence that could be added:

New text: **"The expected yield is parameterised in the likelihood as $S + \sqrt{\mu}I + B$, where S denotes the pure BSM contribution, I denotes the interference between the BSM and SM τ/ν amplitudes, and B denotes the SM background contributions.**

The interference term scales with $\sqrt{\mu}$, since the pure BSM contribution is quadratic while the interference is linear in the BSM amplitude."

page 11, results, third paragraph:

can you quantify satisfactory?

The post-fit background predictions agree with data within approximately 1σ in all VRs except VRW-NonRes, where a deviation of about 2σ is observed.

SK – Sentence modified to clarify:

Old text: Agreement between data and the post-fit background predictions in all VRs is found to be satisfactory.

In the VRW-NonRes region, a deviation of about 2σ from the SM expectation is observed.

New text: **“The post-fit background predictions agree with data within approximately 1σ in all VRs except in the VRW-NonRes region, where a deviation of about 2σ from the SM expectation is observed.”**